

Conformal prediction

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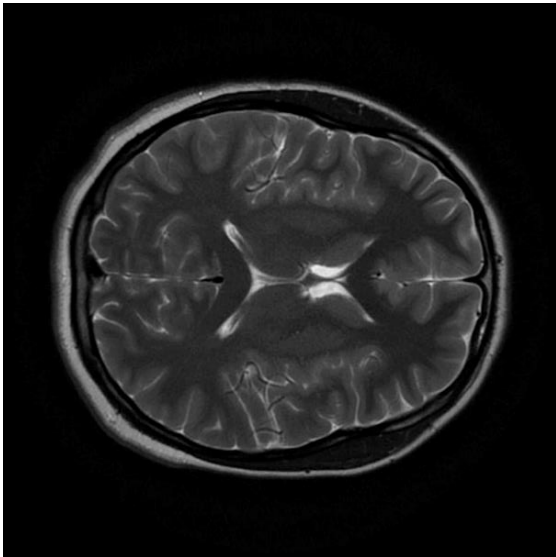
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An important question

How to assess the **confidence** in **a prediction** made by a model?

- A **typical supervised learning task**

x : *MRI images*

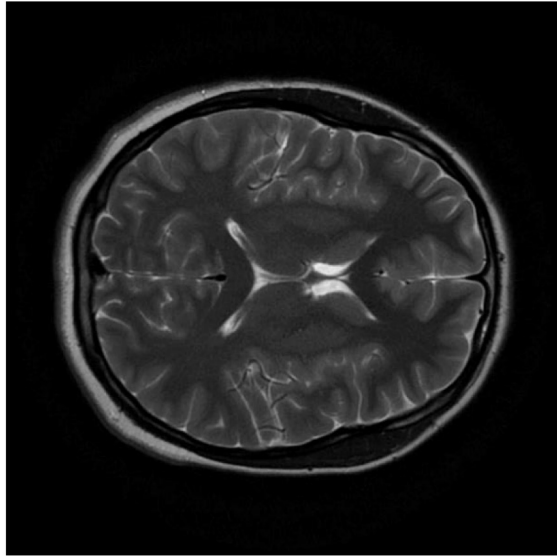


y : *{normal, cancer}*

Usual approach

- Obtain a **training set** of MRI annotated images
- Use these to **learn** a classifier (e.g. a neural net)
- **Evaluate** **accuracy** on the **test set** (e.g. 90%)

x : MRI images



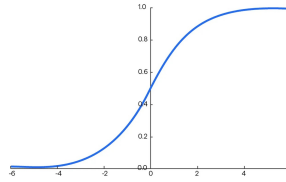
y : {normal, cancer}



Suppose we get 90% accuracy

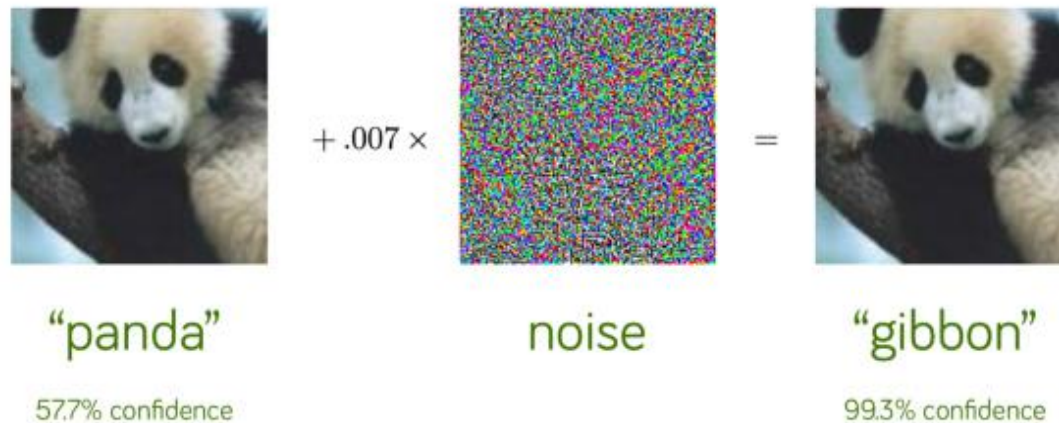
- Is this **enough** for the decision maker (the doctor)?

This **tells us nothing** about the **reliability** or **confidence** for the **current** (yet unseen) **input!**



$$\sigma(\mathbf{z})_j = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}} \quad \forall j \in \{1, \dots, K\}$$

- Often the classifier (e.g. neural net) **output** (e.g. **softmax**) is taken as the **confidence** of the prediction.
- But this often fails miserably



*Explaining and Harnessing Adversarial Examples,
Goodfellow et al, ICLR 2015.*

Softmax likelihoods are generally **poorly calibrated**: they don't reflect the **true probability** of correct classification for the given input

Attempts at **uncertainty quantification**

Deep ensembles

- Train **multiple NNs** using random subsets of data (or same data starting from different random weights)
- Use **predictive distribution** on the input induced by these multiple NNs

But:

- **No correctness guarantees**
- **Computationally expensive**

Lakshminarayanan, Balaji, Alexander Pritzel, and Charles Blundell. "Simple and scalable predictive uncertainty estimation using deep ensembles." NeurIPS 2017.

- What we would like

$$\mathbb{P}(y_{\text{test}} \in \mathcal{C}(x_{\text{test}})) \geq (1 - \alpha)$$



Figure 1: Prediction set examples on Imagenet. We show three progressively more difficult examples of the class *fox squirrel* and the prediction sets (i.e., $\mathcal{C}(X_{\text{test}})$) generated by conformal prediction.

From [ANGELOPOULOS, Anastasios N. et BATES, Stephen. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 2023, vol. 16, no 4, p. 494-591.]

- How could we do that?

Considering the **confidence** in the classifications performed by the system _ the learned hypothesis_ in the **past** to help assess the **confidence** in the **prediction** for the **current input**

The calibration set

1. Define a **confidence** function $c(\mathbf{x}, y)$

E.g. for **classification**

- Let's $h_y(\mathbf{x})$ be the **likelihood** predicted for class y given $\mathbf{x}$
 - E.g. $\text{Softmax}(\mathbf{x}, y)$

2. Randomly draw n labeled examples as the **calibration set** Z

$$Z = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$$

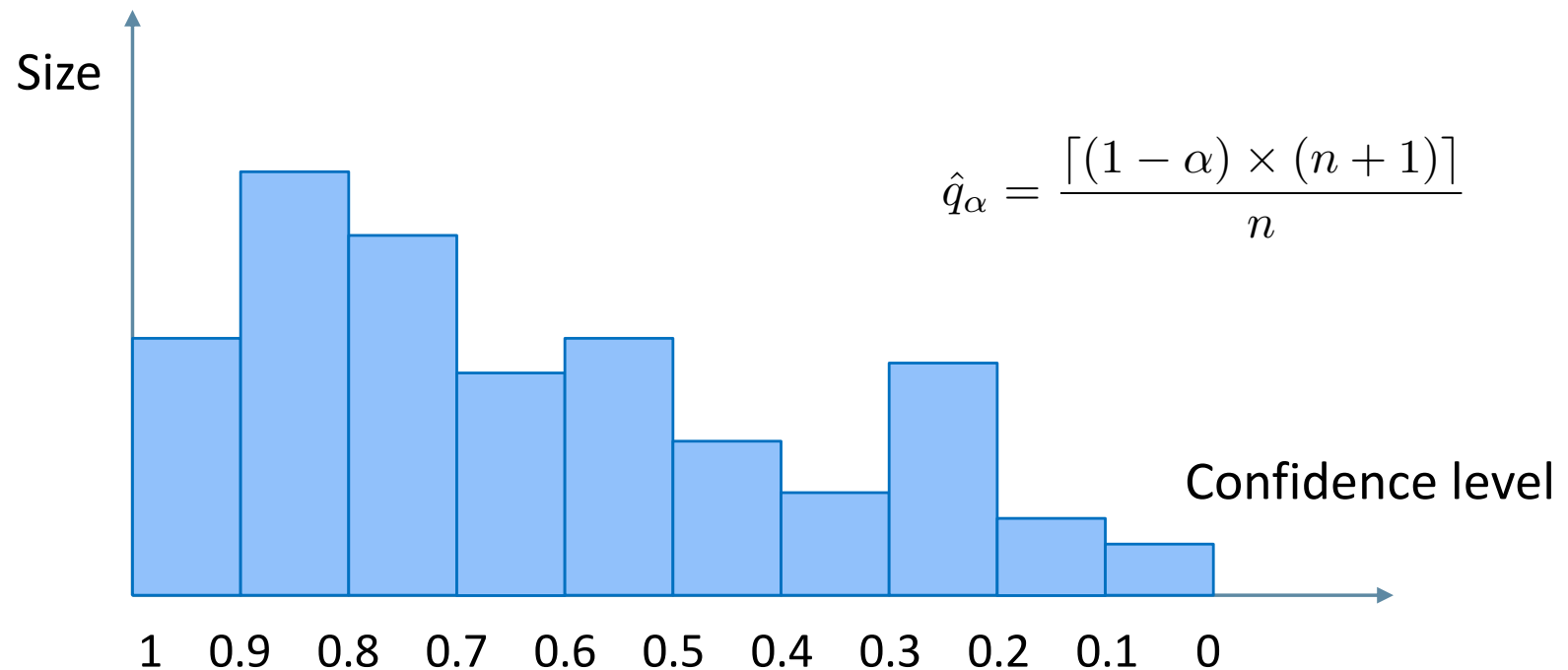
Where y_i is the true class for $\mathbf{x}_i$.

The calibration scores

- Compute the calibration **confidence** levels

$$c(\mathbf{x}_1, y_1), \dots, c(\mathbf{x}_n, y_n)$$

- And compute the **quantiles**



E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.95	0.05	0.10	0.25	0.45	0.25	0.35	0.55	0.10	0.40
$\hat{p}_{\text{tiger}}(X_i)$	0.02	0.90	0.50	0.40	0.30	0.65	0.45	0.15	0.85	0.05
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.40	0.35	0.35	0.10	0.20	0.30	0.05	0.55
	0.95	0.90	0.40	0.35	0.30	0.25	0.20	0.15	0.10	0.05

Confidence in the **true class** for the classifier

$$\hat{q}_\alpha = \lceil (1 - \alpha) \times (n + 1) \rceil$$

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.95	0.05	0.10	0.25	0.45	0.25	0.35	0.55	0.10	0.40
$\hat{p}_{\text{tiger}}(X_i)$	0.02	0.90	0.50	0.40	0.30	0.65	0.45	0.15	0.85	0.05
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.40	0.35	0.35	0.10	0.20	0.30	0.05	0.55
	0.95	0.90	0.40	0.35	0.30	0.25	0.20	0.15	0.10	0.05

$$\alpha = 0.1$$

$$\hat{q}_{0.1} = \lceil (0.9 \times (10 + 1)) \rceil = 10 \quad Q_{0.1} = 0.05$$

Suppose $h(\mathbf{x}) = (0.05, 0.60, 0.35)$

$$\mathcal{C}(\mathbf{x}) = \{\text{dog}, \text{tiger}, \text{cat}\} \quad \text{and} \quad \mathbb{P}(y \in \mathcal{C}(\mathbf{x})) \geq 0.9$$

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.95	0.05	0.10	0.25	0.45	0.25	0.35	0.55	0.10	0.40
$\hat{p}_{\text{tiger}}(X_i)$	0.02	0.90	0.50	0.40	0.30	0.65	0.45	0.15	0.85	0.05
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.40	0.35	0.35	0.10	0.20	0.30	0.05	0.55
	0.95	0.90	0.40	0.35	0.30	0.25	0.20	0.15	0.10	0.05

$$\alpha = 0.5$$

$$\hat{q}_{0.5} = \lceil 0.5 \times (10 + 1) \rceil = 6 \quad Q_{0.5} = 0.30$$

Suppose $h(\mathbf{x}) = (0.05, 0.60, 0.35)$

$$\mathcal{C}(\mathbf{x}) = \{\text{tiger}, \text{cat}\} \quad \text{and} \quad \mathbb{P}(y \in \mathcal{C}(\mathbf{x})) \geq 0.5$$

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.95	0.05	0.10	0.25	0.45	0.25	0.35	0.55	0.10	0.40
$\hat{p}_{\text{tiger}}(X_i)$	0.02	0.90	0.50	0.40	0.30	0.65	0.45	0.15	0.85	0.05
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.40	0.35	0.35	0.10	0.20	0.30	0.05	0.55
	0.95	0.90	0.40	0.35	0.30	0.25	0.20	0.15	0.10	0.05

$$\alpha = 0.9$$

$$\hat{q}_{0.1} = \lceil (0.1 \times (10 + 1)) \rceil = 2$$

$$Q_{0.9} = 0.90$$

Suppose $h(\mathbf{x}) = (0.05, 0.60, 0.35)$

$$\mathcal{C}(\mathbf{x}) = \emptyset \quad \text{and} \quad \mathbb{P}(y \in \mathcal{C}(\mathbf{x})) \geq 0.1$$

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.52	0.45	0.15	0.30	0.25	0.39	0.35	0.35	0.36	0.32
$\hat{p}_{\text{tiger}}(X_i)$	0.45	0.50	0.40	0.28	0.40	0.30	0.27	0.37	0.32	0.35
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.45	0.42	0.35	0.31	0.38	0.28	0.32	0.33
	0.52	0.50	0.45	0.42	0.40	0.39	0.38	0.37	0.36	0.35

Confidence in the **true class** for the classifier

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.52	0.45	0.15	0.30	0.25	0.39	0.35	0.35	0.36	0.32
$\hat{p}_{\text{tiger}}(X_i)$	0.45	0.50	0.40	0.28	0.40	0.30	0.27	0.37	0.32	0.35
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.45	0.42	0.35	0.31	0.38	0.28	0.32	0.33
	0.52	0.50	0.45	0.42	0.40	0.39	0.38	0.37	0.36	0.35

$$\alpha = 0.8$$

$$\hat{q}_{0.8} = \lceil (0.2 \times (10 + 1)) \rceil = 3$$

$$Q_{0.8} = 0.45$$

Suppose $h(\mathbf{x}) = (0.05, 0.60, 0.35)$

$$\mathcal{C}(\mathbf{x}) = \{\text{tiger}\} \quad \text{and} \quad \mathbb{P}(y \in \mathcal{C}(\mathbf{x})) \geq 0.2$$

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.92	0.05	0.02	0.05	0.09	0.84	0.15	0.05	0.78	0.12
$\hat{p}_{\text{tiger}}(X_i)$	0.02	0.90	0.10	0.08	0.86	0.10	0.03	0.80	0.12	0.75
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.88	0.87	0.05	0.06	0.82	0.15	0.10	0.13
	0.92	0.90	0.88	0.87	0.86	0.84	0.82	0.80	0.78	0.75

$$\alpha = 0.1$$

$$\hat{q}_{0.1} = \lceil (0.9 \times (10 + 1)) \rceil = 10$$

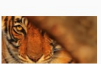
$$Q_{0.1} = 0.75$$

Suppose $h(\mathbf{x}) = (0.05, 0.60, 0.35)$

$$\mathcal{C}(\mathbf{x}) = \emptyset \quad \text{and} \quad \mathbb{P}(y \in \mathcal{C}(\mathbf{x})) \geq 0.9$$

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

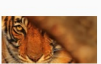
- Calibration set

	dog			tiger				cat		
Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.95	0.90	0.85	0.15	0.15	0.20	0.15	0.15	0.25	0.20
$\hat{p}_{\text{tiger}}(X_i)$	0.02	0.05	0.10	0.60	0.55	0.50	0.45	0.40	0.35	0.45
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.05	0.25	0.30	0.30	0.40	0.45	0.40	0.35
	0.95	0.90	0.85	0.60	0.55	0.50	0.45	0.45	0.40	0.35

Confidence in the **true class** for the classifier

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

	dog			tiger				cat		
Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.95	0.90	0.85	0.15	0.15	0.20	0.15	0.15	0.25	0.20
$\hat{p}_{\text{tiger}}(X_i)$	0.02	0.05	0.10	0.60	0.55	0.50	0.45	0.40	0.35	0.45
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.05	0.25	0.30	0.30	0.40	0.45	0.40	0.35
	0.95	0.90	0.85	0.60	0.55	0.50	0.45	0.45	0.40	0.35

$$\alpha = 0.1$$

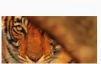
$$\hat{q}_{0.1} = \lceil (0.9 \times (10 + 1)) \rceil = 10 \quad Q_{0.1} = 0.35$$

Suppose $h(\mathbf{x}) = (0.05, 0.60, 0.35)$

$$\mathcal{C}(\mathbf{x}) = \{\text{tiger}, \text{cat}\} \quad \text{and} \quad \mathbb{P}(y \in \mathcal{C}(\mathbf{x})) \geq 0.9$$

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

	dog			tiger				cat		
Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.95	0.90	0.85	0.15	0.15	0.20	0.15	0.15	0.25	0.20
$\hat{p}_{\text{tiger}}(X_i)$	0.02	0.05	0.10	0.60	0.55	0.50	0.45	0.40	0.35	0.45
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.05	0.25	0.30	0.30	0.40	0.45	0.40	0.35
	0.95	0.90	0.85	0.60	0.55	0.50	0.45	0.45	0.40	0.35

$$\alpha = 0.5$$

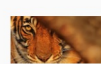
$$\hat{q}_{0.5} = \lceil 0.5 \times (10 + 1) \rceil = 6 \quad Q_{0.5} = 0.55$$

Suppose $h(\mathbf{x}) = (0.05, 0.60, 0.35)$

$$\mathcal{C}(\mathbf{x}) = \{\text{tiger}\} \quad \text{and} \quad \mathbb{P}(y \in \mathcal{C}(\mathbf{x})) \geq 0.5$$

E.g. $Y_i \in \{\text{"dog"}, \text{"tiger"}, \text{"cat"}\}$

- Calibration set

	dog			tiger				cat		
Cal _i										
$\hat{p}_{\text{dog}}(X_i)$	0.95	0.90	0.85	0.05	0.05	0.05	0.05	0.10	0.10	0.15
$\hat{p}_{\text{tiger}}(X_i)$	0.02	0.05	0.10	0.85	0.80	0.75	0.70	0.25	0.30	0.30
$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.05	0.10	0.15	0.20	0.25	0.65	0.60	0.55
	0.95	0.90	0.85	0.85	0.80	0.75	0.70	0.65	0.60	0.55

$$\alpha = 0.1$$

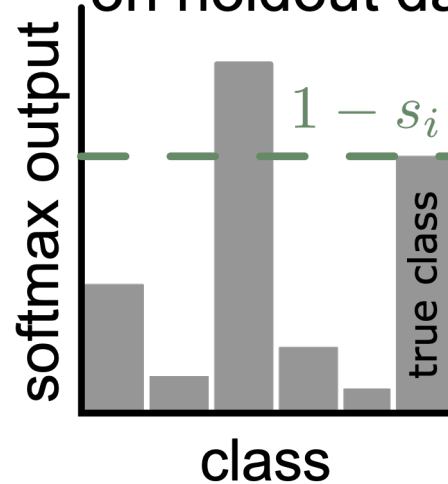
$$\hat{q}_{0.1} = \lceil 0.9 \times (10 + 1) \rceil = 10$$

$$Q_{0.1} = 0.55$$

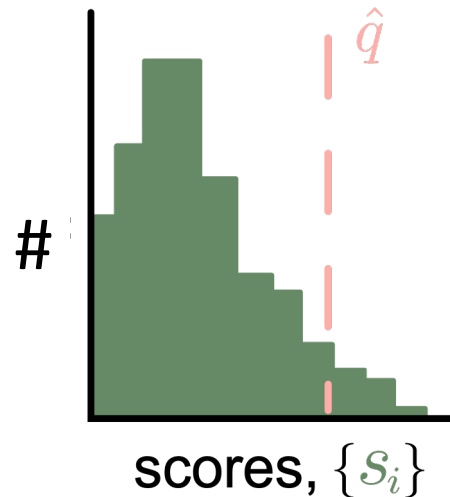
Suppose $h(\mathbf{x}) = (0.05, 0.60, 0.35)$

$$\mathcal{C}(\mathbf{x}) = \{\text{tiger}\} \quad \text{and} \quad \mathbb{P}(y \in \mathcal{C}(\mathbf{x})) \geq 0.9$$

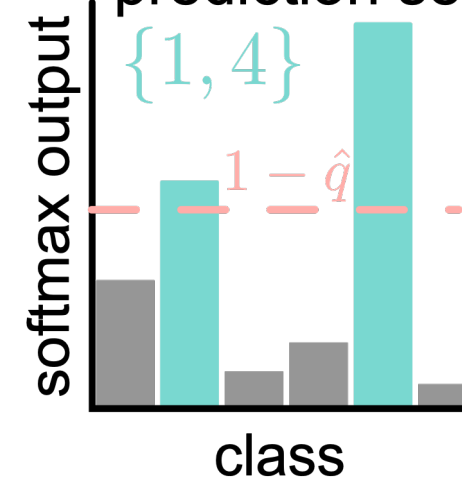
(1) compute scores on holdout data



(2) get quantile



(3) construct prediction set



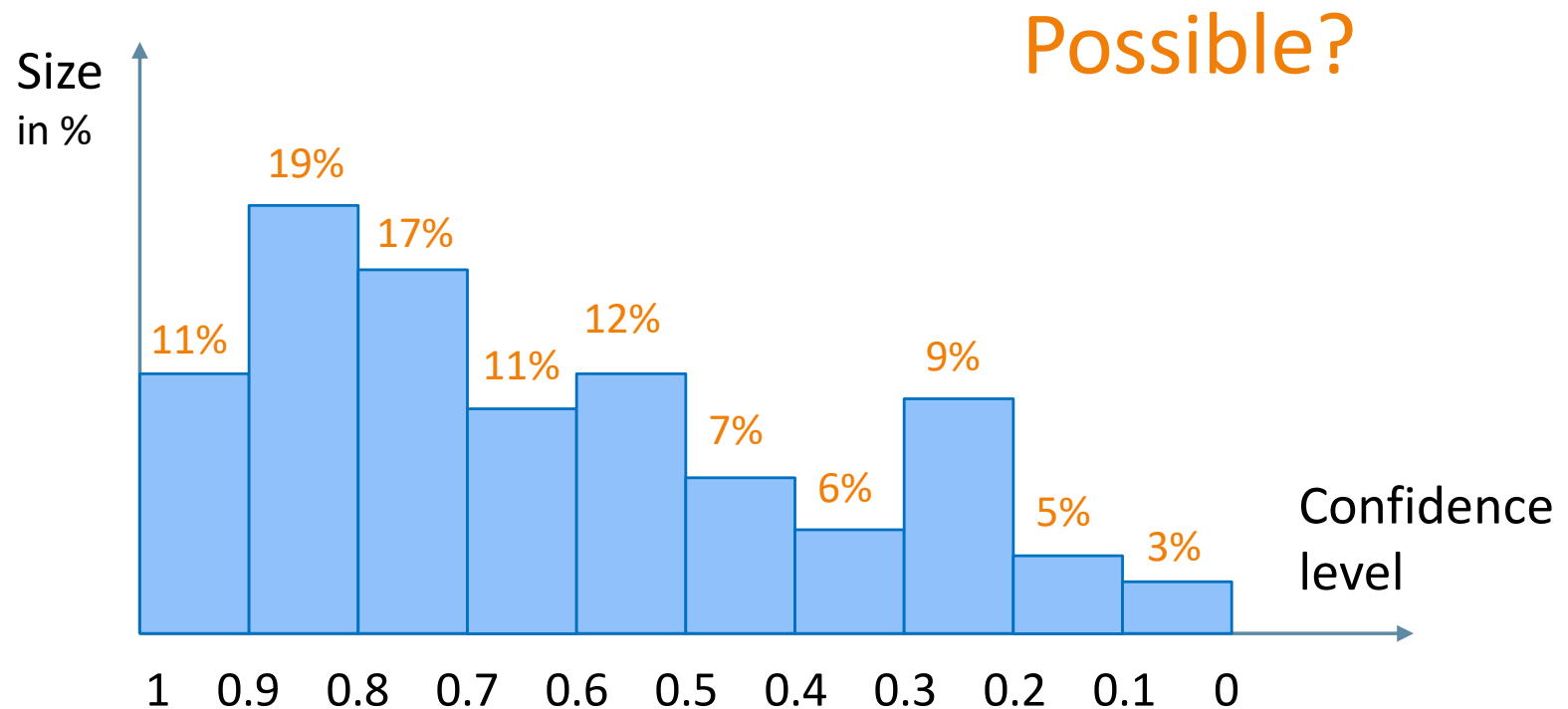
```
# 1: get conformal scores. n = calib_Y.shape[0]
cal_smx = model(calib_X).softmax(dim=1).numpy()
cal_scores = 1-cal_smx[np.arange(n),cal_labels]
# 2: get adjusted quantile
q_level = np.ceil((n+1)*(1-alpha))/n
qhat = np.quantile(cal_scores, q_level, method='higher')
val_smx = model(val_X).softmax(dim=1).numpy()
prediction_sets = val_smx >= (1-qhat) # 3: form prediction sets
```

The calibration scores

- Compute the calibration **confidence** levels

$$c(\mathbf{x}_1, y_1), \dots, c(\mathbf{x}_n, y_n)$$

- And compute the **quantiles**



Take home message

- Provides a measure of uncertainty **for the current input**
- Works
 - with **any distribution**
 - with **any data set size n**
 - with **any predicting model** (for classification or regression)
that provides an estimated confidence in the prediction
- Under assumption that $Z \cup \{(\mathbf{x}_{\text{test}}, y_{\text{test}})\}$ is exchangeable